Article Title

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Abstract

Unmanned Aerial Vehicles (UAVs) rely on the MAVLink protocol for continuous exchange of control commands, telemetry, and mission data between onboard systems and ground control stations. Under real-time operating conditions, a large number of MAVLink messages are generated every second, which can lead to network congestion, increased latency, and delayed delivery of safety-critical messages in bandwidth-constrained environments. In addition, repeated or redundant telemetry messages further increase channel load without providing significant operational benefit.

This paper proposes a machine learning–based message prioritization and congestion mitigation framework for MAVLink-enabled UAV communication systems. A lightweight ML model is deployed at the UAV edge to dynamically assign priority levels to MAVLink messages based on a feature vector $\mathbf{x} = [\boldsymbol{m_t}, \boldsymbol{p_s}, \boldsymbol{f_r}, \boldsymbol{s_u}, \boldsymbol{c_m}]$, where $\boldsymbol{m_t}$ denotes message type, $\boldsymbol{p_s}$ the payload size, $\boldsymbol{f_r}$ the transmission frequency, $\boldsymbol{s_u}$ the system state, and $\boldsymbol{c_m}$ the mission context. The model learns a mapping $\boldsymbol{f}(\mathbf{x}) \rightarrow \boldsymbol{\pi}$, where $\boldsymbol{\pi}$ represents the inferred priority class.

To address congestion, the framework monitors the message arrival rate $\boldsymbol{\lambda}$ and selectively suppresses repeated or low-impact messages when $\boldsymbol{\lambda}$ exceeds a predefined threshold $\boldsymbol{\lambda_{th}}$. Priority-aware queue management ensures that control, navigation, and safety-critical messages are transmitted with minimal delay, while non-critical messages are deferred or discarded during high-load periods. The proposed approach preserves MAVLink backward compatibility by avoiding modifications to the message format or additional protocol overhead.

Experimental evaluation shows that the proposed ML-driven prioritization significantly reduces end-to-end latency and packet loss for high-priority messages,

while maintaining low computational and memory overhead suitable for resource-constrained UAV platforms. The results demonstrate that integrating intelligent prioritization with redundancy elimination improves the robustness and real-time performance of UAV communication systems.

Keywords: UAV Communication Systems, MAVLink, Traffic Prioritization, Edge Intelligence

1 Introduction

1.1 Background and Motivation

Un-Manned Aerial Vehicles are being used in many areas like communication, surveilance, environmental monitoring, disaster management, rescue operations etc that is clearly stated in Ghamari et al. [1]. In many of the cases, one UAV alone cannot manage all the work, so a group of UAV's are deployed and all are collaborated by exchanging messages, and forms UAV Ad Hoc Network (UAANET) Rovira-Sugranes et al. [2] . However, due to high mobility, limited bandwidth, and other dynamic conditions (from Nawaz et al. [3]), some communication packets get dropped/missed which makes reliable communication remains a significant challenge while using UAV's in different situations.

Furthermore, UAANETs differ significantly from traditional mobile ad hoc networks due to their three-dimensional mobility, higher node speeds, and rapidly changing network topology(Amponis et al. [4]). These characteristics make link stability highly unpredictable and routing paths short-lived, leading to frequent route failures and retransmissions. As the number of UAVs participating in the network increases, the communication overhead also grows, which can further strain the limited wireless bandwidth(Yang et al. [5]). Consequently, efficient traffic management becomes essential to prevent network congestion and ensure reliable inter-UAV communication in highly dynamic operating environments.

MAVLink became very popular for transferring the message packets between the UAVs(Cordill et al. [6]). Despite its popularity, MAVLink rely on static or manually configured message transmission rates which may not be suitable for dynamic conditions(Lee [7]) and in some cases congestion may arise in the network where the inflow of the message packets are more than the total bandwidth of the network for communication, as a result some the packets get dropped(Anwekar and Phulre [8]).

In congested network scenarios, packet dropping alone is not an effective solution, as it leads to unnecessary retransmissions, increased delay, and inefficient use of network resources(Anwekar and Phulre [8]). This problem becomes more critical in UAV networks where communication links are often unstable and energy resources are limited. Studies have shown that proactive congestion control techniques, which

regulate traffic flow before buffer overflow occurs, can significantly reduce packet loss and improve overall network performance. In particular, rate-based congestion control approaches are more suitable for UAV networks, as they dynamically adapt the traffic generation rate to current network conditions(Verma et al. [9]).

In addition, UAV communication involves heterogeneous message types such as command and control messages, telemetry data, navigation information, and sensor payload data (Cordill et al. [6]), each having different priority and delay requirements. Treating all messages with the same transmission rate and priority can further aggravate congestion, especially during mission-critical operations where timely delivery of control messages is essential(Ribeiro et al. [10]). Recent studies emphasize that adaptive communication mechanisms which consider message importance and real-time network conditions can significantly improve Quality of Service (QoS) and overall system reliability in UAV Ad Hoc Networks(Buenrostro-Mariscal et al. [11]). Therefore, incorporating congestion-aware and priority-sensitive message rate control mechanisms becomes a crucial requirement for ensuring robust and reliable UAV communication.

Moreover, the dynamic nature of UAV missions implies that network conditions can vary significantly over time due to changes in UAV density, mobility patterns, and communication range. During critical mission phases such as takeoff, formation flying, collision avoidance, or emergency response, the volume of control and telemetry messages may increase sharply, further intensifying network congestion(Mohsan et al. [12]). Static communication strategies are unable to accommodate such temporal traffic variations. Recent studies indicate that adaptive and context-aware communication mechanisms(Rovira-Sugranes et al. [2]), which continuously monitor network conditions and adjust transmission behavior accordingly, are essential for maintaining reliable and scalable UAV communications.

This packet loss can lead to poor Quality of Service (QoS).So these limitations motivates to work and propose adaptive rate control algorithm in UAV Communicaiton.

1.2 Problem Statement

In UAANET, there are different types of messages like command messages, telementry messages, sensor data packets, Location update messages etc. these are being transmitted by static transmission rates. But, in some cases congestion can occur in the UAV Network where the incoming messages are more when compared to the bandwidth of the network and as a result while transmitting the message packets, few will get dropped. As a result the QoS will get effected and failure of task and missions.

Current solutions mainly focus on packet scheduling and messaging routing. while message rate control remains unexplored comparitvely. There is lack of mechanism that dynamically regulates the rate flow based on the congestion.

1.3 Research Objectives and Contributions

The primary objective of this research is to create and evaluate congestion aware rate control mechanism for MAVLink based communication.The proposed approach helps us to adjust the message transmission rate dynamically based on the congestion present in the network which will directly increase the Quality of Service (QoS). This ensure the improvised bandwidth utilization and imporves reliability on MAVLink.

Key Contributions are:

1. Classificaiton of messages using Machine Learning Model and prioritising them.
2. Heuristic rate aware congestion control Algorithm.
3. Message scheduling Algorithm in queue before transmission.

1.4 Organization of the Paper

2 Related Work

2.1 Existing Approaches and Techniques

UAV communication networks have received significant attention in recent years due to their growing adoption in civil, industrial, and defense applications. A comprehensive review by Ghamari et al. [1] discusses UAV communication architectures, spectrum challenges, interference management, and reliability requirements in civil deployments. The study emphasizes that UAV communication must operate under strict latency, reliability, and bandwidth constraints, particularly during mission-critical operations such as disaster monitoring and emergency response.

From a networking perspective, Flying Ad Hoc Networks (FANETs) and UAV Ad Hoc Networks (UAANETs) exhibit unique characteristics including high node mobility, frequent topology changes, intermittent connectivity, and three-dimensional movement. These properties distinguish them from traditional MANETs and VANETs. A cross-layer survey by Amponis et al. [4] highlights the importance of joint optimization across physical, MAC, and network layers to maintain reliable communication in dynamic UAV environments. Similarly, Nawaz et al. [3] identifies bandwidth limitation, packet loss, congestion, and routing instability as major challenges affecting UAV communication performance.

Routing strategies have been extensively studied to address link instability. AI-enabled routing approaches reviewed by Rovira-Sugranes et al. [2] demonstrate that machine learning and reinforcement learning techniques can improve route selection in highly dynamic UAV networks. Adaptive routing designs proposed by Zhang et al. [13] and extended in Zhang et al. [14] use prediction-based mechanisms to estimate link stability and proactively select reliable paths. These approaches improve network-layer reliability; however, they primarily focus on routing optimization rather than application-layer traffic regulation or congestion-aware rate adaptation.

Clustering techniques have also been explored to reduce communication overhead. The weighted and location-based clustering scheme proposed by Yang et al. [5] enhances network scalability by organizing UAVs into structured clusters. Although

clustering reduces routing overhead, it does not directly regulate message transmission rates or prioritize application-layer packets such as MAVLink messages.

At the application layer, MAVLink has emerged as the de facto communication protocol for UAV systems. The survey by Koubâa et al. [15] provides a detailed explanation of MAVLink architecture, message structure, serialization format, and operational characteristics. MAVLink supports periodic and event-driven messages, including heartbeat, telemetry, command, and mission updates. Despite its lightweight design, MAVLink relies on statically configured transmission intervals, which may lead to excessive channel utilization under high-load conditions. Furthermore, Cordill et al. [6] discusses security and privacy issues in UAV systems and highlights that communication overload and congestion can amplify security vulnerabilities by increasing retransmissions and attack surfaces.

To enhance reliability, Lee [7] propose transmitting duplicated control packets across multiple protocols and paths to ensure UAV control robustness. While duplication improves reliability, it may increase network load and aggravate congestion under bandwidth constraints. Similarly, the communication interface manager proposed by Ribeiro et al. [10] enables heterogeneous interface selection to improve throughput and reduce latency. However, these methods focus on path diversity and interface optimization rather than adaptive rate regulation.

Congestion control has been widely studied in wireless sensor and IoT networks. The rate-aware congestion control mechanism proposed by Grover et al. [16] dynamically adjusts packet transmission rates based on network load metrics such as buffer occupancy and packet loss. Their approach effectively reduces packet drops and improves throughput in wireless sensor networks. Likewise, Verma et al. [9] introduce a one-way delay–based adaptive congestion control technique for IoT networks, which proactively regulates traffic before severe congestion occurs. Although these techniques demonstrate the effectiveness of rate-based adaptation, they treat packets uniformly and do not consider message semantics or priority differentiation.

In UAV-specific environments, Anwekar and Phulre [8] analyze congestion control techniques in FANETs and demonstrate that uncontrolled traffic inflow significantly degrades Quality of Service (QoS). The study confirms that congestion-aware rate adaptation can improve reliability and communication stability. Additionally, Buenrostro-Mariscal et al. [11] propose a prioritization-driven cross-layer congestion control mechanism in Internet of Medical Things (IoMT) networks, highlighting the importance of semantic packet prioritization in delay-sensitive environments. However, their work is not tailored to MAVLink-based UAV systems.

Priority-based scheduling approaches have also been proposed in UAV networks. The message scheduling scheme introduced by Li et al. [17] ensures timely delivery of critical UAV messages by assigning predefined priority levels. Similarly, the dynamic super round-based distributed scheduling algorithm proposed by Halder et al. [18] enhances coordination efficiency in multi-UAV systems by organizing communication into structured rounds. More recently, Yang et al. [19] propose an intelligent queue scheduling mechanism for SPMA-based UAV networks, which improves packet handling efficiency at the MAC layer.

Although these scheduling-based methods enhance timeliness and reliability, they primarily operate at MAC or network layers and depend on predefined priority rules. They do not integrate adaptive learning-based classification at the application layer, nor do they dynamically regulate MAVLink transmission rates based on congestion levels.

Furthermore, Mohsan et al. [12] discuss open challenges in UAV systems, emphasizing scalability, congestion management, and adaptive communication as key research directions. The authors highlight that future UAV networks must incorporate intelligent and context-aware mechanisms to handle increasing traffic loads and mission complexity.

Overall, prior research has explored routing optimization, clustering, interface management, congestion control, and priority scheduling independently. However, limited work integrates machine learning–based message classification with congestion-aware rate control specifically for MAVLink-enabled UAV communication systems.

2.2 Comparative Analysis of Prior Studies

Existing literature can be categorized into five major groups: (i) UAV communication surveys and architectural studies, (ii) routing and clustering mechanisms, (iii) reliability enhancement techniques, (iv) congestion control approaches, and (v) priority-based scheduling schemes.

Survey works such as Ghamari et al. [1], Nawaz et al. [3], and Mohsan et al. [12] provide comprehensive insights into UAV communication challenges, including spectrum scarcity, dynamic topology, and congestion. While these studies outline the need for adaptive mechanisms, they do not propose concrete application-layer rate control strategies.

Routing-focused studies [2, 13, 14] improve path stability using prediction and AI-based techniques. Clustering methods such as Yang et al. [5] reduce routing overhead and enhance scalability. However, these approaches operate primarily at the network layer and do not regulate the generation rate of MAVLink messages.

Reliability-enhancing solutions, including multi-path duplication [7] and heterogeneous interface management [10], improve communication robustness but may increase traffic load, potentially worsening congestion.

Congestion control mechanisms proposed for wireless sensor and IoT networks [9, 16] demonstrate the effectiveness of rate adaptation based on congestion metrics. FANET-specific analyses [8] confirm that congestion severely impacts QoS. However, these approaches lack semantic awareness and do not distinguish between critical and non-critical UAV messages.

Priority-based scheduling techniques [17–19] ensure timely transmission of important packets but rely on predefined priority classes and do not integrate machine learning for dynamic classification. Moreover, they focus on scheduling rather than controlling the rate at which messages are generated and transmitted.

The MAVLink survey by Koubâa et al. [15] identifies static transmission configuration as a limitation but does not propose adaptive congestion-aware mechanisms. Security-focused analysis by Cordill et al. [6] further indicates that communication

inefficiencies can exacerbate vulnerabilities, reinforcing the need for intelligent traffic management.

Thus, while substantial progress has been made across routing, scheduling, and congestion control domains, a unified framework that combines machine learning–based message prioritization with congestion-aware rate regulation at the MAVLink application layer remains largely unexplored.

2.3 Research Gaps and Limitations

Despite extensive research, several critical gaps remain in the literature:

- **Lack of Application-Layer Adaptation:** Most routing and clustering techniques [5, 13] operate at lower layers and do not control MAVLink message transmission rates.
- **Static MAVLink Configuration:** MAVLink commonly uses predefined transmission intervals [15], which may lead to congestion under dynamic traffic conditions.
- **Uniform Treatment of Packets:** Congestion control mechanisms [9, 16] treat packets equally without considering semantic importance.
- **Predefined Priority Rules:** Scheduling-based approaches [17, 19] rely on static priority assignments rather than adaptive learning-based classification.
- **Limited Integration of ML with Rate Control:** Although AI-based routing [2] has been explored, integrating machine learning for message-level prioritization combined with congestion-aware rate control remains insufficiently addressed.

Therefore, there is a clear need for an integrated framework that (i) classifies MAVLink messages dynamically using machine learning, (ii) estimates congestion using measurable network parameters, and (iii) regulates transmission rates according to both congestion severity and message priority. The proposed work addresses this gap by introducing a machine learning–based message prioritization mechanism combined with heuristic congestion-aware rate adaptation while maintaining full MAVLink protocol compatibility.

3 System Model / Architecture

3.1 Overall System Architecture

This system architecture has been designed to ease the adaptive, congestion-receptive communication in Unmanned Aerial Vehicle Ad Hoc Networks (UAANETs) and be compatible with MAVLink-based UAV systems. The architecture is modular and separates the message generation, message classification, congestion estimation and adaptive rate control. This form of segregation also allows learning based elements to be flexibly integrated without need of altering the underlying MAVLink protocol. This system operates in a distributed manner and is able to adapt fast to the dynamism of the network conditions a UAV communication environment often experiences.

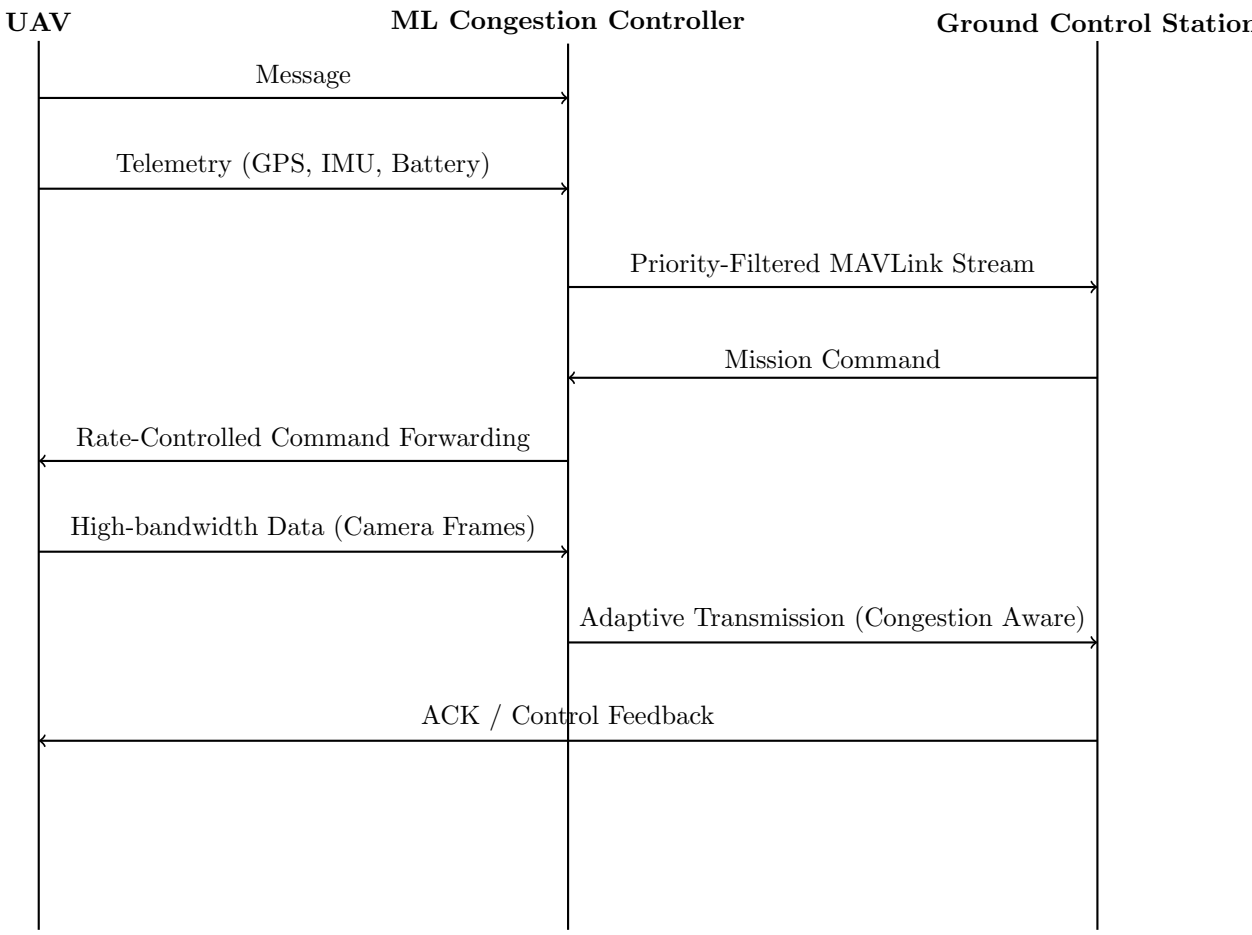

Fig. 1 Time-based MAVLink communication flow with ML-driven congestion-aware rate adaptation between UAV and Ground Control Station.

Different streams of MAVLink messages are produced simultaneously from both UAVs and ground stations, such as command and control messages, telemetry messages, navigation messages and payload sensor messages. These messages vary in a great way with regard to importance, urgency and toleration of delay. In order to deal with such heterogeneity, the architecture proposes a message classification module, which is formed by Machine Learning, that will classify every message to one of the priority classes low, medium, and high according to the message attributes including the type, content, and operational relevance. ML model is trained offline on curated data sets.

Congestion estimation in the proposed architecture is performed using explicit network-level Parameters such as packet loss ratio, queue occupancy, and bandwidth

utilization and they are continuously monitored to quantify the current congestion level of the network. These metrics are combined to derive a normalized congestion factor, which reflects the severity of congestion at a given time. This design choice ensures transparency, low computational overhead, and predictable behavior, which are critical for real-time UAV systems.

Based on the congestion factor and the priority class assigned by the ML model, the adaptive message rate control module dynamically regulates the transmission rates of different message baskets. High-priority messages experience minimal rate reduction to preserve reliable delivery of mission-critical information, while medium- and low-priority messages are progressively throttled as congestion increases. By decoupling message priority classification (ML-based) from congestion estimation (heuristic-based), the architecture achieves a balanced trade-off between intelligence and reliability, resulting in improved Quality of Service (QoS) and robust UAV communication under varying network conditions.

3.2 Network / Communication Model

The network considered in this work is a UAV Ad Hoc Network (UAANET) consisting of multiple UAV nodes communicating over wireless links without fixed infrastructure. The UAVs operate in a shared and bandwidth-limited wireless medium, where network topology changes dynamically due to UAV mobility. Communication links are assumed to be unreliable and subject to packet loss, delay variation, and intermittent connectivity. Depending on mission requirements, communication may occur in single-hop or multi-hop fashion, with each UAV acting as both a data source and a relay node.

The communication between UAVs and the Ground Control Station is based on the MAVLink protocol, which operates at the application layer and typically uses UDP as the underlying transport protocol. MAVLink supports periodic and event-driven message exchanges, making it suitable for real-time UAV communication. In the proposed model, each UAV generates multiple concurrent MAVLink message streams, including heartbeat messages, telemetry data (e.g., GPS, IMU, battery status), navigation information, command and control messages, and high-bandwidth payload data such as camera frames etc.

Each MAVLink message is treated as an independent packet characterized by its message type, size, generation rate, and delay sensitivity. Messages are dynamically classified into three priority categories—high, medium, and low—using a Machine Learning–based classifier. High-priority messages include command and control and safety-critical information, medium-priority messages include telemetry and navigation data, and low-priority messages correspond to bandwidth-intensive or delay-tolerant payload data like temperature updates, battery status updates etc. This classification is performed before transmission and remains independent of the congestion estimation process.

Network congestion is modeled as a condition where the incoming message arrival rate exceeds the available communication capacity. Each UAV monitors local network indicators such as packet loss ratio, queue occupancy, and effective bandwidth utilization to assess the current congestion level. The estimated congestion level is calculated and then used by the adaptive rate control module to regulate the transmission rates of different message priority classes, ensuring reliable communication under dynamic network conditions.

4 Problem Formulation

4.1 Mathematical Model

The UAV communication system is modeled as a dynamic and bandwidth-constrained network in which multiple MAVLink messages are generated and transmitted over a shared wireless channel. Due to the heterogeneous nature of message types and varying network conditions, an adaptive and priority-aware formulation is required.

Let the set of messages generated at a given time instant be defined as:

$$\mathcal{M} = \{m_1, m_2, \ldots, m_K\} \tag{1}$$

where K denotes the total number of messages generated in the system.

Each message $m_i \in \mathcal{M}$ is characterized by a feature vector:

$$\mathbf{x}_i = [m_t, s, p_s, f_r, c, b, m_p, e] \tag{2}$$

where m_t represents the message type, s denotes the source node, p_s is the payload size, f_r is the transmission frequency, c represents the current congestion level, b denotes the battery state, m_p indicates the mission phase, and e is a binary indicator representing emergency conditions.

A machine learning model is employed to classify each message into a priority class:

$$\pi_i = f(\mathbf{x}_i), \quad \pi_i \in \{H, M, L\} \tag{3}$$

where H, M, and L correspond to high, medium, and low priority levels, respectively. This classification introduces semantic awareness into the communication process, enabling differentiated handling of messages based on their operational significance.

The aggregate incoming traffic rate is defined as:

$$\lambda = \sum_{i=1}^{K} f_r^i \tag{4}$$

where f_r^i denotes the generation rate of message m_i. This parameter captures the overall load imposed on the network.

Congestion in the network is said to occur when the incoming traffic rate exceeds the available channel capacity:

$$\lambda > B \tag{5}$$

where B represents the maximum achievable bandwidth. Under this condition, the network experiences buffer accumulation, increased latency, and potential packet loss.

To quantify the severity of congestion, a normalized congestion factor is introduced as:

$$c \in [0, 1] \tag{6}$$

where $c = 0$ corresponds to a congestion-free state and $c = 1$ represents a highly congested network. This normalized representation enables smooth adaptation of transmission rates across varying network conditions.

The transmission rate of each message is dynamically regulated according to the congestion level as follows:

$$R_i^{new} = \begin{cases} R_i^{base}, & \text{if } c \leq c_{th} \\ R_i^{base} \cdot (1 - c(1 - W_i)), & \text{if } c > c_{th} \end{cases} \tag{7}$$

where R_i^{base} denotes the nominal transmission rate of message m_i, R_i^{new} represents the adjusted transmission rate, W_i is the priority weight, and c_{th} is a predefined congestion threshold.

This formulation ensures that, under low congestion conditions, the original transmission behavior is preserved. However, as congestion increases, transmission rates are proportionally reduced based on both the congestion severity and the priority weight. Consequently, high-priority messages undergo minimal rate reduction, while low-priority messages are significantly throttled, thereby preserving critical communication.

The priority weights are defined as:

$$W_i = \begin{cases} 0.9, & \text{if } \pi_i = H \\ 0.6, & \text{if } \pi_i = M \\ 0.3, & \text{if } \pi_i = L \end{cases} \tag{8}$$

which ensures a structured differentiation in transmission control based on message importance.

Furthermore, message scheduling is modeled as a priority-based queueing system:

$$H \rightarrow M \rightarrow L \tag{9}$$

ensuring that high-priority messages are always transmitted before medium- and low-priority messages, thereby improving reliability and timeliness for mission-critical operations.

4.2 Constraints and Parameters

The proposed system is subject to several practical constraints that reflect the limitations of UAV communication environments.

Bandwidth Constraint:

$$\sum_{i=1}^{K} R_i^{new} \leq B \tag{10}$$

This constraint ensures that the total adjusted transmission rate does not exceed the available channel capacity. Maintaining this inequality is essential for system stability, as exceeding the bandwidth would lead to buffer overflow, increased packet loss, and degraded Quality of Service (QoS).

Queue Capacity Constraint:

$$Q_i \leq Q_{\max} \tag{11}$$

where $Q_{\max}$ represents the maximum buffer capacity. This constraint prevents excessive queue buildup, which could otherwise result in packet drops.

Latency Constraint:

$$D_i \leq D_{\max}, \quad \forall i \in H \tag{12}$$

which enforces strict delay requirements for high-priority messages, ensuring timely delivery of critical information.

Energy Constraint:

$$E_{\text{consumed}} \leq E_{\text{available}} \tag{13}$$

reflecting the limited energy resources of UAV systems and the need for energy-efficient communication.

Reliability Constraint:

$$PLR_H \approx 0 \tag{14}$$

which ensures that packet loss for high-priority messages is minimized to maintain safe and reliable UAV operation.

4.3 Optimization / Objective Functions

The primary objective of the proposed system is to maximize the successful transmission of messages while accounting for their relative importance:

$$\max \sum_{i=1}^{K} W_i \cdot T_i \tag{15}$$

where $T_i \in \{0, 1\}$ indicates whether message m_i is successfully transmitted.

In parallel, the system seeks to minimize delay, packet loss, and congestion:

$$\min \left(\sum_{i=1}^{K} D_i + \sum_{i=1}^{K} PLR_i + c \right) \tag{16}$$

Additionally, a composite reward function is defined to guide adaptive decision-making:

$$R = \text{Throughput} + \text{Deadline Satisfaction} - \text{Delay Penalty} - \text{Energy Cost} - \text{Packet Loss} \tag{17}$$

This multi-objective formulation enables a balanced trade-off between throughput maximization, delay minimization, reliability enhancement, and energy efficiency. Unlike conventional approaches that treat all messages uniformly, the proposed formulation integrates machine learning–based prioritization with congestion-aware rate control, making it well-suited for MAVLink-based UAV communication systems.

5 Proposed Methodology

5.1 Overview of the Proposed Approach

The proposed methodology introduces an intelligent and adaptive framework for MAVLink-based UAV communication that integrates machine learning–based message prioritization with congestion-aware rate control and priority-driven scheduling. The primary goal is to ensure reliable and timely delivery of critical messages while efficiently utilizing limited network resources.

The framework consists of three key components: (i) message classification using a trained machine learning model, (ii) congestion estimation using network-level indicators, and (iii) adaptive rate control combined with priority-based scheduling. Unlike conventional approaches that treat all messages uniformly, the proposed system differentiates messages based on their semantic importance and dynamically adjusts their transmission behavior according to network conditions.

Initially, MAVLink messages generated by UAVs are passed through a classification module, where a Random Forest model assigns each message to one of three priority levels: high, medium, or low. This classification is based on features such as message type, payload size, transmission frequency, network congestion, battery level, mission phase, and emergency status.

Subsequently, the system continuously monitors network conditions to estimate congestion levels. Based on the estimated congestion factor and assigned priority, the transmission rate of each message is dynamically regulated. High-priority messages are protected from aggressive rate reduction, while low-priority messages are throttled under high congestion.

Finally, a priority-based scheduling mechanism ensures that messages are transmitted in an ordered manner, giving precedence to critical messages. This integrated approach enhances Quality of Service (QoS), reduces packet loss, and improves overall communication reliability in UAV networks.

5.2 Algorithm / Protocol Design

The proposed framework is designed as an integrated and theoretically grounded communication control mechanism for MAVLink-based UAV systems. It comprises two tightly coupled components: (i) a machine learning–based message prioritization module and (ii) a congestion-aware adaptive rate control and scheduling module. Both components are systematically derived from the mathematical formulation presented in Section **??**, thereby ensuring consistency between analytical modeling and practical implementation.

5.2.1 ML-Based Message Prioritization

The first stage of the framework is responsible for semantic classification of MAVLink messages based on their contextual and operational characteristics. Each message $m_i \in \mathcal{M}$ is represented using the feature vector defined in Eq. (2), which captures key attributes such as message type, payload size, transmission frequency, network congestion, battery status, mission phase, and emergency conditions.

A supervised machine learning model, specifically a Random Forest classifier, is employed to learn the mapping function defined in Eq. (3). The choice of Random Forest is motivated by its robustness to noise, ability to handle heterogeneous feature spaces, and low inference complexity, making it suitable for deployment on resource-constrained UAV platforms.

Based on the learned model, each message is assigned a priority label $\pi_i \in \{H, M, L\}$ corresponding to high, medium, and low priority levels, respectively. This classification enables differentiation between mission-critical and delay-tolerant messages, forming the basis for subsequent transmission control decisions.

Algorithm 1 Machine Learning-Based Message Priority Classification

```
Input: MAVLink message stream M
Output: Priority label π_i for each message
for each message m_i ∈ M do
    Extract feature vector x_i (Eq. 2)
    Predict priority π_i ← f(x_i) (Eq. 3)
    Assign π_i ∈ {H, M, L}
end for
```

The output of this stage is a semantically enriched message stream in which each message is annotated with a priority label. These labels serve as critical inputs to the congestion-aware transmission control mechanism described next.

5.2.2 Congestion-Aware Rate Control and Scheduling

The second stage of the framework dynamically regulates message transmission rates and scheduling decisions based on both network congestion and message priority. This component is derived directly from the rate adaptation model defined in Eq. (7) and the priority weighting scheme in Eq. (8).

For each incoming message, a priority weight W_i is assigned according to its classified priority. Simultaneously, the network congestion level is estimated using the normalized congestion factor defined in Eq. (6). Based on these parameters, the transmission rate of each message is adjusted adaptively.

Under low congestion conditions ($c \leq c_{th}$), messages are transmitted at their base rates to maintain standard communication behavior. However, when congestion exceeds the threshold ($c > c_{th}$), transmission rates are reduced according to the priority-aware rate control function. This ensures that high-priority messages are preserved, while non-critical traffic is progressively throttled.

Algorithm 2 Priority-Aware Congestion Control and Scheduling

Input: Labeled message stream $\mathcal{M}$ with priority π_i
Output: Priority-aware transmission schedule
for each message $m_i \in \mathcal{M}$ **do**
    Assign weight W_i based on Eq. (8)
    Estimate congestion factor c (Eq. 6)
    if $c \leq c_{th}$ **then**
        $R_i^{new} \leftarrow R_i^{base}$
    else
        $R_i^{new} \leftarrow R_i^{base} \cdot (1 - c(1 - W_i))$ (Eq. 7)
    end if
    Enforce minimum transmission constraints for emergency messages
    Insert message into priority queue $\mathcal{Q}_{\pi_i}$
end for
while channel capacity is available **do**
    Transmit messages according to priority order (Eq. 9)
end while

This adaptive mechanism ensures that the aggregate transmission rate remains within the available bandwidth, thereby satisfying the constraint defined in Eq. (10). By proactively regulating message rates, the system effectively prevents queue overflow and reduces packet loss.

5.2.3 Integration of ML and Rate Control

The overall protocol operates as a sequential and interdependent pipeline, where the output of the machine learning module directly influences the behavior of the congestion control mechanism. Specifically, the priority label π_i obtained from Eq. (3) is mapped to a corresponding weight W_i using Eq. (8). This weight governs the degree of rate adaptation applied in Eq. (7).

During periods of low congestion, all messages are transmitted at their nominal rates, ensuring minimal deviation from standard MAVLink operation. In contrast, under high congestion conditions, the rate control mechanism selectively reduces transmission rates in a priority-aware manner. High-priority messages, characterized by larger weights, experience minimal rate degradation, whereas low-priority messages undergo significant throttling.

In addition, the priority-based scheduling policy defined in Eq. (9) ensures that critical messages are always transmitted ahead of less important traffic. This coordinated interaction between prioritization, rate adaptation, and scheduling enables the system to maintain reliable communication even under constrained network conditions.

Overall, the proposed framework establishes a direct correspondence between the mathematical formulation and its algorithmic realization. The machine learning component provides intelligent and context-aware prioritization, while the congestion control module ensures efficient utilization of network resources, reduced packet loss, and improved Quality of Service (QoS) in UAV communication systems.

5.3 Workflow and Operational Phases

The operation of the proposed system can be divided into four major phases:

1. **Message Generation Phase:** UAVs generate MAVLink messages including telemetry, navigation, control, and sensor data at varying rates.
2. **Classification Phase:** Each message is processed by the machine learning model, which assigns a priority label based on contextual and operational features.
3. **Congestion Estimation Phase:** The system continuously monitors network parameters such as packet loss ratio, queue occupancy, and bandwidth utilization to compute a congestion factor.
4. **Adaptive Transmission Phase:** Based on the congestion level and message priority:
   - Transmission rates are adjusted dynamically
   - Messages are queued according to priority
   - Scheduling ensures high-priority messages are transmitted first

This phased workflow ensures that the system adapts in real time to changing network conditions while maintaining efficient communication.

5.4 Computational Complexity Analysis (optional)

The computational complexity of the proposed system is analyzed in terms of its major components:

- **Machine Learning Classification:** The Random Forest classifier has a prediction complexity of $\mathcal{O}(T \cdot d)$, where T is the number of trees and d is the depth of each tree. Since inference is performed at runtime, it remains computationally efficient for UAV edge devices.
- **Rate Control Mechanism:** The rate adjustment formula involves constant-time operations, resulting in $\mathcal{O}(1)$ complexity per message.
- **Queue Management and Scheduling:** Insertion into priority queues and selection for transmission are performed in $\mathcal{O}(1)$ time using separate queues for each priority level.
- **Overall Complexity:** For K messages, the total complexity is:

$$\mathcal{O}(K \cdot T \cdot d)$$

The proposed approach is lightweight and suitable for real-time deployment in resource-constrained UAV systems, as it balances computational efficiency with intelligent decision-making.

6 Implementation Details

6.1 Simulation Environment and Tools

The simulation is implemented in Python 3.10 using NumPy for numerical computations, Pandas for dataset management, Matplotlib for result visualization. A fixed

random seed (42) is used throughout to ensure reproducibility of all stochastic channel jitter draws.

6.2 Dataset Description

The simulation dataset contains 500 rows, each representing one discrete time step of a MAVLink-equipped UAV communicating within a UAANET. There are different source nodes are present, reflecting a realistic swarm topology with bidirectional ground-to-air traffic. The four mission phases — TakeOff, Cruise, Swarm, and Landing are distributed broadly which represents different stages of the status of UAV, with no single phase dominating, ensuring the simulation covers diverse operational contexts.

The six MAVLink message types, their distribution in the dataset, assigned priority labels, and base transmission rates used in the simulation. Priority labels are assigned by the ML classifier based on message semantics: Alert and Command messages are safety-critical (High); Navigation messages carry route information (Medium); Heartbeat, Telementry and Video messages are either periodic or bandwidth-intensive, it is classified based on other features also(Low or High based on operational context).

6.3 Parameter Settings

All simulation parameters are fixed prior to execution and held constant for both FIFO and proposed algorithm runs. Table I lists the complete parameter set with values and descriptions.

Parameter	**Symbol**	**Value**	**Description**
Network Capacity	C	25 pkt/step	Nominal channel throughput ceiling
Queue Limit	$Q_{\max}$	60 packets	Global maximum buffer size before drop
Channel Jitter	ϵ	$\pm 10\%$	Random capacity variation per step
Low Congestion Threshold	η_L	0.40	Below this: no rate change applied
High Congestion Threshold	η_H	0.50	At or above this: rate control active
Emergency Emergency	—	+0.10	Added to η when emergency_flag = 1
Low-Battery Emergency	—	+0.08	Added to η when battery $< 20\%$
Priority Weights	W	H=0.9 / M=0.6 / L=0.3	Rate Adjustment factor
Simulation Steps	N	500	Total dataset rows processed
Random Seed	—	42	Fixed for full reproducibility

Table 1 Simulation Parameters Considered

7 Performance Evaluation and Results

7.0.1 ML Model Performance Evaluation

The performance of the proposed Random Forest-based message prioritization model is evaluated using standard classification metrics, including Accuracy, Precision, Recall, and F1-score. The evaluation is conducted on a test set obtained using a 75:25 train-test split over a dataset of 500 samples.

Table 2 Overall Performance Metrics of the Random Forest Model

Metric	Value
Accuracy	97.60%
Precision (Weighted)	97.84%
Recall (Weighted)	97.60%
F1-Score (Weighted)	97.47%

The results indicate that the model achieves high classification accuracy with well-balanced precision and recall, making it suitable for reliable priority assignment in UAV communication systems.

Table 3 Class-wise Performance Evaluation

Class	Precision	Recall	F1-Score	Support
High (H)	1.00	1.00	1.00	87
Low (L)	0.90	1.00	0.95	27
Medium (M)	1.00	0.73	0.84	11

A detailed class-wise analysis reveals that high-priority messages are perfectly classified, ensuring reliable identification of critical communication. Low-priority messages also exhibit strong performance with high recall, enabling effective deprioritization under congestion.

However, medium-priority messages show relatively lower recall (0.73), indicating minor misclassification with adjacent classes. Despite this, the precision remains high, confirming that predicted medium-priority labels are reliable.

Table 4 Average Performance Metrics

Average Type	Precision	Recall	F1-Score
Macro Average	0.97	0.91	0.93
Weighted Average	0.98	0.98	0.97

The weighted average metrics confirm the robustness of the model across all classes, despite class imbalance. Overall, the Random Forest classifier provides accurate and consistent priority predictions, forming a reliable foundation for the subsequent congestion-aware rate control and scheduling mechanism. The high accuracy in identifying high-priority messages is particularly critical for maintaining Quality of Service (QoS) in UAV communication systems.

7.1 Performance Metrics

The Analysis of the proposed heuristic Rate control algorithm is done on the basis of four quantitative performance metrics computed over all 500 simulation steps. they are:

- **Average Throughput:** The number of packets successfully delivered per time step, measuring how effectively the channel capacity is utilized in the experiment.
- **Average Packet Loss:** The mean number of packets dropped per step due to queue overflow, directly measuring how well each algorithm prevents buffer saturation under congestion.
- **Total Packets Delivered:** The aggregate count of all successfully transmitted packets over the 500-step simulation, providing a cumulative measure of overall delivery performance.
- **Packet Delivery Ratio(PDR):** the ratio of total delivered packets to total attempted packets, expressed as a percentage. This is the primary QoS metric. the formulae used to calculate the PDR is

$$\text{PDR (\%)} = \frac{\text{Total_Delivered}}{\text{Total_Delivered} + \text{Total_Dropped}} \times 100$$

7.2 Experimental Scenarios

Two algorithms for rate control transmission are evaluated under identical conditions, the same 500-step dataset, the same effective congestion sequence, and the same channel capacity model with the same random seed is used.

- **FIFO Baseline:** A single shared queue processes all messages in arrival order. Packets enter the queue at the raw send frequency recorded in the dataset (mean 25.55 Hz, max 49.93 Hz) without any rate adjustment. The channel capacity degrades with congestion per the same model (Eq. above), and any queue occupancy exceeding 60 packets is dropped. This represents the current state of practice for static MAVLink deployments.
- **Proposed ML-Based Priority-Aware Congestion Control:** Unlike the FIFO baseline, the proposed algorithm calculate the congestion and the queue size availability before trying to add the packet into the queue. Before a message enters into queue, its transmission rate is dynamically adjusted by the congestion-aware rate

control formula based on the current effective congestion level and the message's assigned priority weight.

7.3 Results and Analysis

The comparision between the proposed heuristic model and FIFO Model is evaluated and results are tabulated below.

Metric	**FIFO (Baseline)**	**Proposed Algorithm**	**Improvement**
Avg Throughput (pkt/step)	17.996	17.013	Almost similar
Avg Packet Loss (pkt/step)	7.443	1.134	Very less packet loss
Total Packets Delivered	8,997.8	8,506.4	comparitively less packets delivered
Total Packets Dropped	3,721.6	567.0	very few packets dropped
Packet Delivery Ratio (%)	70.74%	93.75%	+23.01% improved

Table 5 Performance Comparison: FIFO Baseline vs. Proposed Algorithm ($N = 500$ steps)

Metric	**FIFO Baseline**	**Proposed Algorithm**
PDR (%)	70.74%	93.75%
Avg Loss / step	7.443 pkt	1.134 pkt
Total Dropped	3,721.6	567.0
Total Delivered	8,997.8	8,506.4
Avg Throughput	17.996 pkt/step	17.013 pkt/step

Table 6 FIFO Baseline vs. Proposed Algorithm Performance

Table [3] presents a quantitative comparison of the FIFO baseline and the proposed algorithm across five performance metrics over the 500-step simulation. The most significant result is the Packet Delivery Ratio, which improves from 70.74% under FIFO to 93.75% with the proposed algorithm — a gain of 23.01 percentage. This improvement is directly attributed to the 84.8% reduction in total dropped packets, falling from 3,721.6 to 567 across the full simulation run.

8 Discussion

8.1 Key Observations

8.2 Practical Implications

8.3 Limitations of the Study

9 Conclusion and Future Work

9.1 Summary of Findings

9.2 Future Research Directions

Several promising directions remain for extending this work. First, hardware validation on physical flight controllers such as ArduPilot and PX4 with live MAVLink traffic streams will verify whether the performance gains observed in simulation hold under real-world conditions including OS scheduling latency, radio retransmissions, and hardware interrupts.

Second, multi-UAV swarm evaluation involving multiple concurrent communication links sharing a common wireless channel will assess the scalability of the proposed framework and quantify the impact of inter-link interference on priority scheduling effectiveness.

Third, dynamic threshold adaptation will replace the fixed congestion thresholds η_L (lower congestion limit) = 0.40 and η_H(upper congestion limit)= 0.50 with an Artificial intelligence (AI) estimator that continuously adjusts both values based on rolling channel utilization and queue occupancy metrics, allowing the algorithm to self-tune across diverse and changing mission environments.

References

[1] Ghamari, M., Rangel, P., Mehrubeoglu, M., Tewolde, G.S., Sherratt, R.S.: Unmanned aerial vehicle communications for civil applications: A review. IEEE Access **10**, 102492–102531 (2022)

[2] Rovira-Sugranes, A., Razi, A., Afghah, F., Chakareski, J.: A review of ai-enabled routing protocols for uav networks: Trends, challenges, and future outlook. Ad Hoc Networks **130**, 102790 (2022)

[3] Nawaz, H., Ali, H.M., Laghari, A.A.: Uav communication networks issues: A review: H. nawaz et al. Archives of Computational Methods in Engineering **28**(3), 1349–1369 (2021)

[4] Amponis, G., Lagkas, T., Sarigiannidis, P., Vitsas, V., Fouliras, P., Wan, S.: A survey on fanet routing from a cross-layer design perspective. Journal of Systems Architecture **120**, 102281 (2021)

[5] Yang, X., Yu, T., Chen, Z., Yang, J., Hu, J., Wu, Y.: An improved weighted and

location-based clustering scheme for flying ad hoc networks. Sensors **22**(9), 3236 (2022)

[6] Cordill, B., Fang, D., Xu, S.: A comprehensive survey of security and privacy in uav systems. IEEE Access (2025)

[7] Lee, W.: Enabling reliable uav control by utilizing multiple protocols and paths for transmitting duplicated control packets. Sensors **21**(9), 3295 (2021)

[8] Anwekar, D., Phulre, S.: Analysis of congestion control techniques to improve qos and frequent communication in fanet. In: 2023 World Conference on Communication & Computing (WCONF), pp. 1–7 (2023). IEEE

[9] Verma, L.P., Kumar, G., Khalaf, O.I., Wong, W.-K., Hamad, A.A., Rawat, S.S.: Adaptive congestion control in iot networks: Leveraging one-way delay for enhanced performance. Heliyon **10**(22) (2024)

[10] Ribeiro, L.M.B., Müller, I., Buss Becker, L.: Communication interface manager for improving performance of heterogeneous uav networks. Sensors **21**(13), 4255 (2021)

[11] Buenrostro-Mariscal, R., Santana-Mancilla, P.C., Montesinos-López, O.A., Vazquez-Briseno, M., Nieto-Hipolito, J.I.: Prioritization-driven congestion control in networks for the internet of medical things: A cross-layer proposal. Sensors **23**(2), 923 (2023)

[12] Mohsan, S.A.H., Othman, N.Q.H., Li, Y., Alsharif, M.H., Khan, M.A.: Unmanned aerial vehicles (uavs): Practical aspects, applications, open challenges, security issues, and future trends. Intelligent service robotics **16**(1), 109–137 (2023)

[13] Zhang, M., Dong, C., Yang, P., Tao, T., Wu, Q., Quek, T.Q.: Adaptive routing design for flying ad hoc networks. IEEE Communications Letters **26**(6), 1438–1442 (2022)

[14] Zhang, M., Cheng, H., Yang, P., Dong, C., Zhao, H., Wu, Q., Quek, T.Q.: Adaptive routing design for flying ad hoc networks: A joint prediction approach. IEEE Transactions on Vehicular Technology **73**(2), 2593–2604 (2023)

[15] Koubâa, A., Allouch, A., Alajlan, M., Javed, Y., Belghith, A., Khalgui, M.: Micro air vehicle link (mavlink) in a nutshell: A survey. IEEE Access **7**, 87658–87680 (2019)

[16] Grover, A., Kumar, R.M., Angurala, M., Singh, M., Sheetal, A., Maheswar, R.: Rate aware congestion control mechanism for wireless sensor networks. Alexandria Engineering Journal **61**(6), 4765–4777 (2022)

[17] Li, J., Chen, M., Dai, F., Wang, H.: Prioritizing-based message scheduling for reliable unmanned aerial vehicles ad hoc network. International Journal of

Performability Engineering **14**(9), 2021 (2018)

[18] Halder, S., Ghosal, A., Conti, M.: Dynamic super round-based distributed task scheduling for uav networks. IEEE Transactions on Wireless Communications **22**(2), 1014–1028 (2022)

[19] Yang, K., Xu, C., Qiao, G., Zhong, J., Zhang, X.: Intelligent queue scheduling method for spma-based uav networks. Drones **9**(8), 552 (2025)